Trajectory Visualization System Using Computer Vision

Project Report

Prepared for GitHub Submission

Abstract

This project focuses on developing a trajectory visualization system using computer vision techniques. The system estimates motion from a video stream and visualizes the movement trajectory in real time. OpenCV-based visual tracking methods are used to detect and track features between frames. The generated trajectory provides a basic understanding of motion estimation and visual SLAM concepts.

Objectives

• Develop a real-time trajectory visualization system.
• Estimate motion using image processing techniques.
• Visualize camera movement dynamically.
• Understand the fundamentals of visual odometry and SLAM.
• Create a software-only prototype suitable for future hardware integration.

Technologies Used

The project uses the following technologies and tools:

• Python Programming Language
• OpenCV Library
• NumPy
• Matplotlib (optional visualization)
• Laptop Webcam / Video Input
• GitHub for version control and project hosting

System Workflow

1. Capture video frames from the webcam or recorded video.
2. Convert frames to grayscale for processing.
3. Detect visual features using OpenCV feature detectors.
4. Track feature movement between consecutive frames.
5. Estimate motion direction and displacement.
6. Draw and update the trajectory path dynamically.
7. Display output visualization in real time.

Implementation Details

The implementation uses OpenCV for image acquisition and feature tracking. Consecutive video frames are compared to estimate motion. Key points are tracked using optical flow methods such as Lucas-Kanade Optical Flow. The displacement values are accumulated and plotted on a trajectory map. Noise reduction and smoothing techniques are applied to reduce drift and instability.

Applications

• Autonomous Robotics
• Visual SLAM Research
• Indoor Navigation Systems
• Drone Motion Tracking
• Educational Computer Vision Projects
• Robotics and AI-based Localization Systems

Results

The system successfully visualizes motion trajectories based on camera movement. Real-time tracking is achieved using standard computer vision techniques. Minor drift may occur due to accumulated estimation errors, but the prototype demonstrates the core concept of trajectory estimation effectively.

Future Scope

Future improvements may include:

• Integration with LiDAR sensors such as TF-Luna.
• Sensor fusion using IMU data.
• Deep learning-based feature extraction.
• Full Visual SLAM implementation.
• ROS integration for robotics applications.
• 3D trajectory reconstruction and mapping.

Conclusion

This project demonstrates a basic trajectory visualization system using computer vision and motion estimation techniques. It provides a strong foundation for future work in robotics, SLAM, autonomous systems, and AI-driven navigation technologies. The project also helps in understanding practical implementation concepts related to real-time visual tracking systems.

Project Components

Component	Purpose
Python	Main programming language
OpenCV	Image processing and tracking
NumPy	Matrix and numerical operations
GitHub	Project hosting and version control
Webcam	Video input source